Modelagem do Sistema de Injeção

Lista de Abreviações e Siglas

Abreviação	**Significado**
AC	Água de Resfriamento - Área Classificada
AD	Água Destilada
ADO	Água para Diluição de Óleo
AF	Água Fresca
AFDO	Água Fresca de Diluição de Óleo
AI	Anti-incrustante
AL	Alcalino
AM	Água do Mar
ANC	Água de Resfriamento - Área Não Classificada
AQ	Água Quente
B	Bomba
BB	Bomba Booster
BC	Bomba de Captação
BD	Biodispersante
BI	Biocida
BP	Bypass
CAP	Captação
DR	Dreno
DAM	Distribuição de Água do Mar
DAR	Distribuição de Água de Resfriamento
DES	Desaeradora
DOS	Dosadora
EC	Eletrocloradora
FF	Filtro Fino
FG	Filtro Grosso
FT	Filtro
GI	Gás Inerte
HC	Hipoclorito
INJ	Injeção
INT	Interconexões
LIM	Limpeza
LAV	Estações de Lavagem
LINHAAF	Linha de Água Fresca
LINHAAD	Linha de Água Destilada
LQ	Limpeza Química
LQDO	Limpeza Química - Diluição de Óleo
MIS	Mistura
MK	Tanque Makeup
NaHSO3	Bissulfito de Sódio
OB	Overboard
OD	Open Drain
P	Permeado
R	Retorno
SEQ	Sequestrante

Abreviação	**Significado**
SK	Skid
SL	Safe Location
SO	Sequestrante de Oxigênio
SR	Sistema de Resfriamento
TA, TB, TC, TD, TE	Trem A, B, C, D, E
TC	Trocador de Calor
TOT	
TQ	Tanque
URS	Unidade de Remoção de Sulfatos
UTI	Utilidades
VAC	
VE	Vessel
VENT	Vent

Variáveis calculadas

$F_{4,out,AD,UTI}$

$F_{4,out,AF,UTI}$

$F_{5,out,DAM,UTI}$

$F_{6,out,AC,UTI}$

1 Captação de Água do Mar

1.1 Eletrocloradora

$$F_{7,in,AM} = F_{7,in,BB_ABC} + F_{7,out,AM_DR} \tag{1}$$

$$F_{7,in,BB_ABC} = F_{7,out,BB_ABC} \tag{2}$$

$$\boldsymbol{F_{7,out,BB_ABC}} = F_{7,out,BB_ABC,DR} + F_{7,out,BB_ABC,MIS_1} \tag{3}$$

$$F_{7,out,BB_ABC,MIS_1} = F_{7,in,SR} + F_{7,in,EC} \tag{4}$$

$$F_{7,in,EC} = F_{7,in,EC_1} + F_{7,in,EC_2} \tag{5}$$

$$F_{7,in,EC_1} = F_{7,out,EC_1} + F_{7,out,EC_1,DR} \tag{6}$$

$$F_{7,in,EC_2} = F_{7,out,EC_2} + F_{7,out,EC_2,DR} \tag{7}$$

$$F_{7,in,TQ_HC,EC} = F_{7,out,EC_1} + F_{7,out,EC_2} \tag{8}$$

$$F_{7,out,EC_1,DR} + F_{7,out,EC_2,DR} = F_{7,out,EC_12,DR} \tag{9}$$

$$F_{7,out,AM_DR} + F_{7,out,BB_ABC,DR} + F_{7,out,EC_12,DR} + \boldsymbol{F_{7,out,SR,DR}} = F_{7,EC,DR} \tag{10}$$

1.2 Tanque de Hipoclorito

$$F_{7,in,B_ABC} + F_{7,out,TQ_HC,DR} = F_{7,in,TQ_HC,EC} + F_{7,in,TQ_HC,B_ABC} \tag{11}$$

$$F_{7,in,B_ABC} = F_{7,in,TQ_HC,B_ABC} + \boldsymbol{F_{7,in,INJ,B_ABC}} + F_{7,out,B_ABC,DR} \tag{12}$$

$$F_{7,out,HC,DR} = F_{7,out,TQ_HC,DR} + F_{7,out,B_ABC,DR} \tag{13}$$

1.3 Captação de Água do Mar e Injeção de Hipoclorito

$$F_{7,in,CAP,INJ_HC1} = F_{7,in,BC_A,HC1} + F_{7,out,INJ_HC1,DR} \tag{14}$$

$$F_{7,out,BC_A} = F_{7,in,BC_A,HC1} + F_{7,in,BC_A,AM} \tag{15}$$

$$F_{7,in,CAP,INJ_HC2} = F_{7,in,BC_B,HC2} + F_{7,out,INJ_HC2,DR} \tag{16}$$

$$F_{7,out,BC_B} = F_{7,in,BC_B,HC2} + F_{7,in,BC_B,AM} \tag{17}$$

$$F_{7,in,CAP,INJ_HC3} = F_{7,in,BC_C,HC3} + F_{7,out,INJ_HC3,DR} \tag{18}$$

$$F_{7,out,BC_C} = F_{7,in,BC_C,HC3} + F_{7,in,BC_C,AM} \tag{19}$$

$$F_{7,in,CAP,INJ_HC4} = F_{7,in,BC_D,HC4} + F_{7,out,INJ_HC4,DR} \tag{20}$$

$$F_{7,out,BC_D} = F_{7,in,BC_D,HC4} + F_{7,in,BC_D,AM} \tag{21}$$

$$F_{7,in,CAP,INJ_HC5} = F_{7,in,BC_E,HC5} + F_{7,out,INJ_HC5,DR} \tag{22}$$

$$F_{7,out,BC_E} = F_{7,in,BC_E,HC5} + F_{7,in,BC_E,AM} \tag{23}$$

$$F_{7,out,BC_A} + F_{7,out,BC_B} + F_{7,out,BC_C} + F_{7,out,BC_D} = F_{6,in,F_ABCD,BC_ABCD} \tag{24}$$

$$\begin{aligned} F_{7,out,INJ_HC,DR} = F_{7,out,INJ_HC1,DR} + F_{7,out,INJ_HC2,DR} + F_{7,out,INJ_HC3,DR} \\ + F_{7,out,INJ_HC4,DR} + F_{7,out,INJ_HC5,DR} \end{aligned} \tag{25}$$

$$F_{7,out,INJ_TC,DR} = F_{7,out,INJ_HC,DR} + F_{6,out,F_ABCDE,BC_OB} \tag{26}$$

2 Distribuição de Água

$$F_{5,in,DAM} = F_{6,out,TC,DAM_LAV} + F_{6,out,F_ABCDE,DAM} \tag{27}$$

$$F_{5,in,DAM} = F_{5,in,FG,DAM} + F_{8,in,DES_SK,DAM} + F_{5,out,DAM,UTI} \tag{28}$$

2.1 Filtros - Distribuição

$$\begin{aligned} F_{6,in,F_ABCD,BC_ABCD} + F_{6,in,F_E,BC_E} &= F_{6,in,TC,F_ABCD} \\ &+ F_{6,out,F_ABCD,DAM} + F_{6,out,F_ABCD,VAC_1} + F_{6,out,F_ABCD,VAC_2} \\ &+ F_{6,in,ANC,F_ABCDE} + F_{6,out,F_ABCDE,BC_OB} + F_{6,out,F_ABCDE,FG_OB} \\ &F_{6,out,F_ABCDE,FG_OD} \end{aligned} \tag{29}$$

$$F_{6,in,F_A} = F_{6,out,F_A} + F_{6,out,F_A,OD} \tag{30}$$

$$F_{6,in,F_B} = F_{6,out,F_B} + F_{6,out,F_B,OD} \tag{31}$$

$$F_{6,in,F_C} = F_{6,out,F_C} + F_{6,out,F_C,OD} \tag{32}$$

$$F_{6,in,F_D} = F_{6,out,F_D} + F_{6,out,F_D,OD} \tag{33}$$

$$F_{6,in,F_E} = F_{6,out,F_E} + F_{6,out,F_E,OD} \tag{34}$$

2.2 Trocadores de Calor

$$\begin{aligned} F_{6,in,TC,F_ABCD} + F_{6,in,TC,INT_INJ_BB} + F_{6,in,TC,AC} &= F_{6,out,TC,DAR} \\ &+ F_{6,in,AC,TC} + F_{6,out,TC,DAM_LAV} + F_{8,in,INT_INJ_B,TC} \\ &+ F_{6,out,TC,OB1} + F_{6,out,TC,OB2} + F_{6,out,TC,AC_OB} \end{aligned} \tag{35}$$

$$F_{6,in,TC_A1} + F_{6,in,TC_A2} = F_{6,out,TC_A} + F_{6,out,TC_A,OB} \tag{36}$$

$$F_{6,in,TC_B1} + F_{6,in,TC_B2} = F_{6,out,TC_B} + F_{6,out,TC_B,OB} \tag{37}$$

$$F_{6,in,TC_C1} + F_{6,in,TC_C2} = F_{6,out,TC_C} + F_{6,out,TC_C,OB} \tag{38}$$

$$F_{6,in,TC_D1} + F_{6,in,TC_D2} = F_{6,out,TC_D} + F_{6,out,TC_D,OB} \tag{39}$$

$$F_{6,in,TC_E1} + F_{6,in,TC_E2} = F_{6,out,TC_E} + F_{6,out,TC_E,OB} \tag{40}$$

2.3 Água de Refrigeração - AC

$$F_{6,in,AC,DAR_1} + F_{6,in,AC,DAR_2} + F_{6,in,AC,TC} + F_{6,in,AC,AF_MK} + F_{6,in,AC,GI} = F_{6,in,TC,AC} + F_{6,out,AC,UTI} + F_{6,out,AC,ANC_OD1} + F_{6,out,AC,TC_OB} + F_{6,out,AC,ANC_OD2} + F_{6,out,AC,SL_1} + F_{6,out,AC,SL_2} + F_{6,out,AC,VE_OB} + F_{6,out,AC,B_AB_OB} + F_{6,out,AC,F_OB} \quad (41)$$

$$F_{6,in,B_A1} = F_{6,out,B_A1} + F_{6,out,B_A1,OD} \quad (42)$$

$$F_{6,in,B_B1} = F_{6,out,B_B1} + F_{6,out,B_B1,OD} \quad (43)$$

$$F_{6,in,B_C1} = F_{6,out,B_C1} + F_{6,out,B_C1,OD} \quad (44)$$

$$F_{6,in,B_A2} = F_{6,out,B_A2} + F_{6,out,B_A2,OD} + F_{6,out,B_A2,OB} \quad (45)$$

$$F_{6,in,B_B2} = F_{6,out,B_B2} + F_{6,out,B_B2,OD} + F_{6,out,B_B2,OB} \quad (46)$$

$$F_{6,in,F} = F_{6,out,F} + F_{6,out,F,OB} \quad (47)$$

$$F_{6,in,VE1_1} + F_{6,in,VE1_2} + F_{6,in,VE1_3} = F_{6,in,SL_1} + F_{6,in,SL_2} + F_{6,out,VE1} + F_{6,out,VE1,OB} \quad (48)$$

$$F_{6,in,VE2} = F_{6,out,VE2_1} + F_{6,out,VE2_2} \quad (49)$$

2.4 Água de Refrigeração - ANC

$$F_{6,in,ANC,F_ABCDE} + F_{6,in,ANC,B_AQ} + F_{6,in,ANC,DAR} = F_{6,in,TC,ANC} + F_{6,out,ANC,AC_OB} + F_{6,out,ANC,DAR} + F_{6,out,ANC,B_AQ} \quad (50)$$

$$F_{6,in,B_A} = F_{6,out,B_A} + F_{6,out,B_A,OD_OB} \quad (51)$$

$$F_{6,in,B_B} = F_{6,out,B_B} + F_{6,out,B_B,OD_OB} \quad (52)$$

$$F_{6,in,B_C} = F_{6,out,B_C} + F_{6,out,B_C,OD_OB} \quad (53)$$

$$F_{6,in,TC_A1} + F_{6,in,TC_A2} = F_{6,out,TC_A1} + F_{6,out,TC_A2} + F_{6,out,TC_A,OD_OB} \tag{54}$$

$$F_{6,in,TC_B1} + F_{6,in,TC_B2} = F_{6,out,TC_B1} + F_{6,out,TC_B2} + F_{6,out,TC_B,OD_OB} \tag{55}$$

$$F_{6,in,VE} = F_{6,out,VE} + F_{6,out,VE,OD_OB} \tag{56}$$

$$F_{6,in,TQ} = F_{6,out,TQ_1} + F_{6,out,TQ_2} \tag{57}$$

3 Tratamento da Água

3.1 Filtros Grossos

$$F_{1,in,FG,DAM} + F_{1,in,FG,BI_SK1} = F_{1,in,FF,FG} + F_{1,out,FG_ABC,OD} + F_{1,out,FG_ABC,OB} \tag{58}$$

$$F_{1,in,FG_A} = F_{1,out,FG_A} + F_{1,out,FG_A,OB} + F_{1,out,FG_A,OD} \tag{59}$$

$$F_{1,in,FG_B} = F_{1,out,FG_B} + F_{1,out,FG_B,OB} + F_{1,out,FG_B,OD} \tag{60}$$

$$F_{1,in,FG_C} = F_{1,out,FG_C} + F_{1,out,FG_C,OB} + F_{1,out,FG_C,OD} \tag{61}$$

$$F_{1,in,FF,FG} = F_{1,out,FG_A} + F_{1,out,FG_B} + F_{1,out,FG_C} \tag{62}$$

3.2 Filtros Finos

$$F_{1,in,FF,FG} + F_{1,in,FF,BI_SK1} + F_{1,in,FF,NaHSO3_SK3} + \boldsymbol{F_{1,in,FF,AI_SK2}} = F_{1,out,FF,AF} + F_{1,in,BB,FF} + F_{1,out,FF_ABC,OB} + F_{1,out,FF_ABC,OD} \tag{63}$$

$$F_{1,in,FF_A} = F_{1,out,FF_A} + F_{1,out,FF_A,OB} + F_{1,out,FF_A,OD} \tag{64}$$

$$F_{1,in,FF_B} = F_{1,out,FF_B} + F_{1,out,FF_B,OB} + F_{1,out,FF_B,OD} \tag{65}$$

$$F_{1,in,FF_C} = F_{1,out,FF_C} + F_{1,out,FF_C,OB} + F_{1,out,FF_C,OD} \tag{66}$$

$$F_{1,in,BB,FF} = F_{1,out,FF_A} + F_{1,out,FF_B} + F_{1,out,FF_C} \tag{67}$$

3.3 Bomba Booster

$$F_{1,in,BB,FF} = F_A + F_B + F_C + F_D + F_E + F_{1,in,URS_BP,BB} + F_{1,in,TA_OB} + F_{1,in,FF_OD} \tag{68}$$

$$\boldsymbol{F_{1,out,BB,URS}} = F_{1,in,URS_BP,BB} + F_{1,in,TA_OB} \tag{69}$$

$$F_{1,in,BB,FF} = F_{1,in,BB_A} + F_{1,in,BB_B} \tag{70}$$

$$F_{1,in,BB_A} = F_{1,out,BB_A} + F_{1,out,BB_A,OD} \tag{71}$$

$$F_{1,in,BB_B} = F_{1,out,BB_B} + F_{1,out,BB_B,OD} \tag{72}$$

3.4 Sistema de Membranas da URS

3.4.1 Reconciliação

$$F = \boldsymbol{R} + P \tag{73}$$

$$P = P_A + P_B + P_C + P_D + P_E \tag{74}$$

$$P_A = \boldsymbol{P_{a,A}} + \boldsymbol{P_{b,A}} + \boldsymbol{P_{c,A}} \tag{75}$$

$$P_B = \boldsymbol{P_{a,B}} + \boldsymbol{P_{b,B}} + \boldsymbol{P_{c,B}} \tag{76}$$

$$P_C = \boldsymbol{P_{a,C}} + \boldsymbol{P_{b,C}} + \boldsymbol{P_{c,C}} \tag{77}$$

$$P_D = \boldsymbol{P_{a,D}} + \boldsymbol{P_{b,D}} + \boldsymbol{P_{c,D}} \tag{78}$$

$$P_E = \boldsymbol{P_{a,E}} + \boldsymbol{P_{b,E}} + \boldsymbol{P_{c,E}} \tag{79}$$

$$\boldsymbol{R} = \boldsymbol{R_{c,A}} + \boldsymbol{R_{c,B}} + \boldsymbol{R_{c,C}} + \boldsymbol{R_{c,D}} + \boldsymbol{R_{c,E}} \tag{80}$$

$$F = F_A + F_B + F_C + F_D + F_E \tag{81}$$

$$F_A = \boldsymbol{R_{a,A}} + \boldsymbol{P_{a,A}} + R_{b,A} + \boldsymbol{P_{b,A}} \tag{82}$$

$$F_B = \boldsymbol{R_{a,B}} + \boldsymbol{P_{a,B}} + R_{b,B} + \boldsymbol{P_{b,B}} \tag{83}$$

$$F_C = \boldsymbol{R_{a,C}} + \boldsymbol{P_{a,C}} + R_{b,C} + \boldsymbol{P_{b,C}} \tag{84}$$

$$F_D = \boldsymbol{R_{a,D}} + \boldsymbol{P_{a,D}} + R_{b,D} + \boldsymbol{P_{b,D}} \tag{85}$$

$$F_E = \boldsymbol{R_{a,E}} + \boldsymbol{P_{a,E}} + R_{b,E} + \boldsymbol{P_{b,E}} \tag{86}$$

$$R_{b,A} = F_{c,A} - \boldsymbol{R_{a,A}} = \boldsymbol{R_{c,A}} + \boldsymbol{P_{c,A}} - \boldsymbol{R_{a,A}} \tag{87}$$

$$R_{b,B} = F_{c,B} - \boldsymbol{R_{a,B}} = \boldsymbol{R_{c,B}} + \boldsymbol{P_{c,B}} - \boldsymbol{R_{a,B}} \tag{88}$$

$$R_{b,C} = F_{c,C} - \boldsymbol{R_{a,C}} = \boldsymbol{R_{c,C}} + \boldsymbol{P_{c,C}} - \boldsymbol{R_{a,C}} \tag{89}$$

$$R_{b,D} = F_{c,D} - \boldsymbol{R_{a,D}} = \boldsymbol{R_{c,D}} + \boldsymbol{P_{c,D}} - \boldsymbol{R_{a,D}} \tag{90}$$

$$R_{b,E} = F_{c,E} - \boldsymbol{R_{a,E}} = \boldsymbol{R_{c,E}} + \boldsymbol{P_{c,E}} - \boldsymbol{R_{a,E}} \tag{91}$$

$$F_{8,in,DES,URS} = P + F_{1,in,URS_BP,BB} \tag{92}$$

3.4.2 Monitoramento por Estágios

Algumas variáveis representam o mesmo ponto físico, mas recebem nomes diferentes em cada contexto (estágio / banco). Adota-se um nome canônico único para cada ponto, de forma a evitar redundância no sistema de equações. A Tabela **??** apresenta o mapeamento adotado.

Nome canônico	Nomes equivalentes
PT_{RF}	$PT_{1R} = PT_{2F} = PT_{cF}$
PT_R	$PT_{cR} = PT_{2R}$
PT_P	$PT_{2P} = PT_{cP}$
PDI_{c2}	$PDI_c = PDI_2$

$$S_1 = \frac{\dfrac{\boldsymbol{P_a} + \boldsymbol{P_b}}{\boldsymbol{A}}}{\boldsymbol{PT_{RF}} - PT_{1P} - \boldsymbol{\Delta\pi}} \tag{93}$$

$$S_2 = \frac{\dfrac{\boldsymbol{P_c}}{\boldsymbol{A}}}{\boldsymbol{PT_R} - PT_P - \boldsymbol{\Delta\pi}} \tag{94}$$

$$\boldsymbol{IDF_1} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_{RF}} - PT_{1P} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_a} + \boldsymbol{P_b}}{\boldsymbol{A}} \tag{95}$$

$$\boldsymbol{IDF_2} = \boldsymbol{L_p} \cdot (\boldsymbol{PT_R} - PT_P - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_c}}{\boldsymbol{A}} \tag{96}$$

$$\boldsymbol{PDI_1} = \boldsymbol{PT_{1F}} - \boldsymbol{PT_{RF}} \tag{97}$$

$$\boldsymbol{PDI_{c2}} = \boldsymbol{PT_{RF}} - \boldsymbol{PT_R} \tag{98}$$

3.4.3 Monitoramento por Bancos

$$S_a = \frac{\dfrac{\boldsymbol{P_a}}{\boldsymbol{A}}}{PT_{aR} - PT_{aP} - \boldsymbol{\Delta\pi}} \tag{99}$$

$$S_b = \frac{\dfrac{\boldsymbol{P_b}}{\boldsymbol{A}}}{PT_{bR} - PT_{bP} - \boldsymbol{\Delta\pi}} \tag{100}$$

$$S_c = \frac{\dfrac{\boldsymbol{P_c}}{\boldsymbol{A}}}{\boldsymbol{PT_R} - PT_P - \boldsymbol{\Delta\pi}} \tag{101}$$

$$IDF_a = \boldsymbol{L_p} \cdot (PT_{aR} - PT_{aP} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_a}}{\boldsymbol{A}} \tag{102}$$

$$IDF_b = \boldsymbol{L_p} \cdot (PT_{bR} - PT_{bP} - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_b}}{\boldsymbol{A}} \tag{103}$$

$$IDF_c = \boldsymbol{L_p} \cdot (PT_R - PT_P - \boldsymbol{\Delta\pi}) - \frac{\boldsymbol{P_c}}{\boldsymbol{A}} \tag{104}$$

$$PDI_a = PT_{aF} - PT_{aR} \tag{105}$$

$$PDI_b = PT_{bF} - PT_{bR} \tag{106}$$

$$\boldsymbol{PDI_{c2} = PT_{RF} - PT_R} \tag{107}$$

3.5 Injeção de Biocida

$$F_{1,in,TQ_1,TOT_BI} = F_{1,in,FG,BI_SK1} + F_{1,in,FF,BI_SK1} + F_{1,out,TQ_1,LIM_OD} + F_{1,out,SK1,LIM_OD} \tag{108}$$

$$F_{1,in,TQ_1} = F_{1,out,TQ_1,LIM_OD} + F_{1,in,SK1} \tag{109}$$

$$F_{1,in,SK1} = F_{1,in,FG,BI_SK1} + F_{1,in,FF,BI_SK1} + F_{1,out,SK1,LIM_OD} \tag{110}$$

$$\boldsymbol{F_{1,out,SK1}} = F_{1,in,FG,BI_SK1} + F_{1,in,FF,BI_SK1} \tag{111}$$

3.6 Injeção de Anti-incrustante

$$F_{3,in,TQ_2,TOT_AI} = F_{1,in,FF,AI_SK2} + F_{3,out,TQ_2,LIM_OD} + F_{3,out,SK2,LIM_OD} \tag{112}$$

$$F_{3,in,TQ_2} = F_{3,out,TQ_2,LIM_OD} + F_{3,in,SK2} \tag{113}$$

$$F_{3,in,SK2} = \boldsymbol{F_{1,in,FF,AI_SK2}} + F_{3,out,SK2,LIM_OD} \tag{114}$$

3.7 Injeção de Bissulfito de Sódio

$$F_{3,in,TQ_3,TOT_NaHSO3} = F_{1,in,FF,NaHSO3_SK3} + F_{4,in,LIM,NaHSO3_SK3} + F_{3,out,TQ_3,LIM_OD} + F_{3,out,SK3,LIM_OD} \tag{115}$$

$$F_{3,in,TQ_3} = F_{3,out,TQ_3,LIM_OD} + F_{3,in,SK3} \tag{116}$$

$$F_{3,in,SK3} = F_{1,in,FF,NaHSO3_SK3} + F_{4,in,LIM,NaHSO3_SK3} + F_{3,out,SK3,LIM_OD} \tag{117}$$

$$\boldsymbol{F_{3,out,SK3}} = F_{1,in,FF,NaHSO3_SK3} + F_{4,in,LIM,NaHSO3_SK3} \tag{118}$$

3.8 Sistema de Limpeza

$$F_{4,in,LIM,AD} + F_{4,in,LIM,NaHSO3_SK3} = F_{4,in,MIS_1} + F_{4,in,MIS_2} \tag{119}$$

$$F_{4,in,RLQ,TABCDE} + F_{4,in,RLQ,AFDO} = F_{4,in,MIS_3} + F_{4,in,MIS_4} \tag{120}$$

$$F_{4,in,TQ_A,AL} + F_{4,in,TQ_A,AC} + F_{4,in,MIS_1} + F_{4,in,MIS_3} = F_{4,out,TQ_A} + F_{4,out,TQ_A,MIS_OD} \tag{121}$$

$$F_{4,in,TQ_B,AL} + F_{4,in,TQ_B,AC} + F_{4,in,MIS_2} + F_{4,in,MIS_4} + F_{4,in,TQ_B,B_ABC} = F_{4,out,TQ_B} + F_{4,out,TQ_B,MIS_OD} \tag{122}$$

$$F_{4,out,TQ_A} + F_{4,out,TQ_B} = F_{4,in,B_A} + F_{4,in,B_B} + F_{4,in,B_C} \tag{123}$$

$$F_{4,in,B_A} = F_{4,out,B_A} \tag{124}$$

$$F_{4,in,B_B} = F_{4,out,B_B} \tag{125}$$

$$F_{4,in,B_C} = F_{4,out,B_C} \tag{126}$$

$$F_{4,out,B_A} + F_{4,out,B_B} + F_{4,out,B_C} = F_{4,in,TQ_B,B_ABC} + F_{4,in,F_3} \tag{127}$$

$$F_{4,in,F_3} = \boldsymbol{F_{4,out,F_3}} + F_{4,out,F_3,ADO_OB} \tag{128}$$

$$F_{4,out,F_3} = F_{4,in,LQ_TABCDE,FT3} + F_{4,in,LQDO_TA,FT3} + F_{4,in,LQDO_TB,FT3} \quad (129)$$

$$F_{4,in,LINHAAF} = F_{4,out,AF,UTI} + F_{4,in,ANC,AF} + F_{4,in,UTI_AQ,AF} + F_{4,in,AF_MK,AF} \quad (130)$$

$$F_{4,in,AF_MK,AF} = F_{4,out,AF_MK,B_AQ} + F_{6,in,AC,AF_MK} + F_{4,out,AF_MK,OB} \quad (131)$$

$$F_{4,in,LINHAAD} + F_{4,in,SO} = F_{4,out,AD,UTI} + F_{4,in,LIM,AD} \quad (132)$$

4 Injeção de Água no Poço

4.1 Desaeração a Vácuo

$$F_{8,in,DES,URS} + F_{8,in,DES,BI} = F_{8,in,DES,URS_BI} + F_{8,in,INJ,URS_BI} \quad (133)$$

$$F_{8,out,DES_G} = F_{8,out,DES_G1} + F_{8,out,DES_G2} + F_{8,out,VENT} \quad (134)$$

$$F_{8,in,DES,URS_BI} + F_{8,in,DES,INJ} + F_{8,in,DES,SE} = F_{8,out,DES_1} + F_{8,out,DES_2} + F_{8,out,DES_G} \quad (135)$$

$$F_{8,in,INJ,DES} + F_{8,out,DES_1,OB} = F_{8,out,DES_1} + F_{8,in,INJ,URS_BI} + F_{8,in,INJ,BI} + F_{8,in,INJ,SE} + F_{8,in,INJ,BD} \quad (136)$$

$$F_{8,out,SK,FF_OB} = F_{8,in,SK,G_1} + F_{8,in,SK,G_2} + F_{8,in,DES_SK,DAM} + F_{8,in,SK,AR} \quad (137)$$

4.2 Bombeamento de Água de Injeção

$$F_{8,in,INJ_BB,DES} = F_{8,in,BB_A} + F_{8,in,BB_B} + F_{8,in,BB_C} \quad (138)$$

$$F_{8,in,BB_A} = F_{8,out,BB_A} + F_{8,out,BB_A,OD} \quad (139)$$

$$F_{8,in,BB_B} = F_{8,out,BB_B} + F_{8,out,BB_B,OD} \quad (140)$$

$$F_{8,in,BB_C} = F_{8,out,BB_C} + F_{8,out,BB_C,OD} \quad (141)$$

$$F_{8,out,BB_A} + F_{8,out,BB_B} + F_{8,out,BB_C} = F_{6,in,TC,INT_INJ_BB} + F_{8,out,BB,URS} \quad (142)$$

$$F_{8,in,INT_INJ_B,TC} = F_{8,in,B_A,TC} + F_{8,in,B_B,TC} \quad (143)$$

$$\boldsymbol{F_{8,in,B_A,TC}} + F_{8,in,B_A,DAR} = F_{8,out,B_A} + F_{8,out,B_A,OD} + F_{8,out,B_A,URS_OB} + F_{8,out,B_A,ANC} \quad (144)$$

$$\boldsymbol{F_{8,in,B_B,TC}} + F_{8,in,B_B,DAR} = F_{8,out,B_B} + F_{8,out,B_B,OD} + F_{8,out,B_B,URS_OB} + F_{8,out,B_B,ANC} \quad (145)$$

$$\boldsymbol{F_{8,out,B_AB}} = F_{8,out,B_A} + F_{8,out,B_B} \quad (146)$$

$$\boldsymbol{F_{8,out,B_AB}} = F_{8,in,AI,IJN} + F_{8,in,ADO,INJ} \quad (147)$$